

Basic Principles of Medicine 2
Module: Digestion and Metabolism
DLA

**Histology of the Gastrointestinal
System – Overview**

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Objectives

Describe the distinct four layers characteristic of the alimentary canal.	SOM.MK.I.BPM2.3.DM.1.HCB.1185
Differentiate between Meissner's and Auerbach's plexuses based on their microscopic structure, function, localization and origins.	SOM.MK.I.BPM2.3.DM.1.HCB.1186
Compare serosa vs. adventitia.	SOM.MK.I.BPM2.3.DM.1.HCB.1187

Digestive System

Consist of:

1. The **alimentary canal** which begins at the mouth and ends at the anal canal. **Contains the gastrointestinal tract** which transport ingested food, modifies it for easy absorption at different sites
2. Associated organs such as **digestive glands**, tongue and teeth

The functions of the alimentary canal are therefore:

- Physical and chemical digestion
- Absorption
- Excretion of digestive waste
- Secretion of hormones, digestive juices and electrolytes
- Physical barrier – prevents entry of noxious substances, antigens and pathological organisms
- Immunological protection

Origin of the alimentary canal:

- The epithelium of the alimentary canal and its associated glands is derived from the endoderm off the primitive gut tube
- This is true for all areas of the canal except epithelia of the oral cavity and anal canal which are derived from ectoderm
- The connective and muscular tissue components are derived from mesoderm. The peritoneal linings which covers some of the organs is specifically from the lateral plate mesoderm

Alimentary Canal – Gastrointestinal tract

The gastrointestinal tract extends from proximal esophagus to anal canal. It has the same basic structure throughout its length. The tract is organized in four distinct layers. Below these layers are listed based on their proximity to the lumen of the canal and color coded in both illustrations

1. Mucosa
2. Submucosa
3. Muscularis Externa
4. Serosa/Adventitia

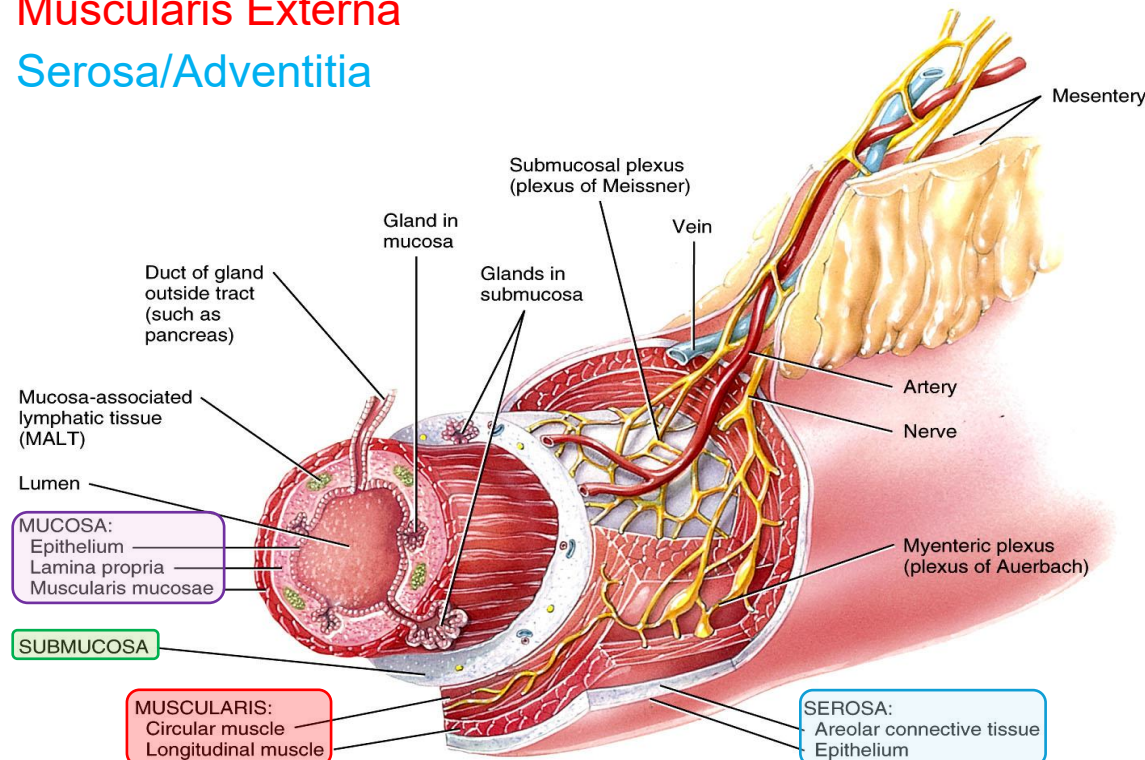
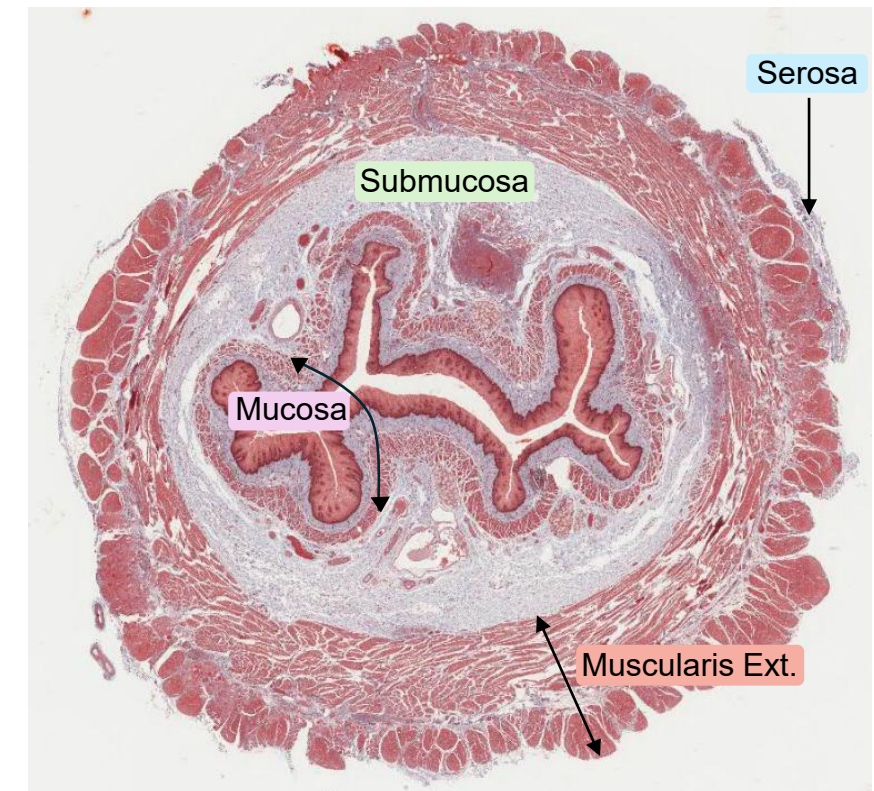


Figure 24.02 Tortora - PAP 12/e
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Now let's see what this looks like in this trichrome stained histology slide



Let's take a closer look at the Mucosa

The mucosa varies considerably from region to region however three distinct layers is present:

1. **Epithelium**
2. **Lamina Propria** contains loose connective tissue, blood & lymph vessels, mucosal glands, gut-associated lymphatic tissue (GALT)
3. **Muscularis Mucosae** forms the boundary between mucosa and submucosa. It is usually formed by 2 layers of smooth muscle, inner circular and outer longitudinal → **look out for the exception in lecture**. Contraction produces movement of the mucosal components and changes the surface area for absorption or secretion

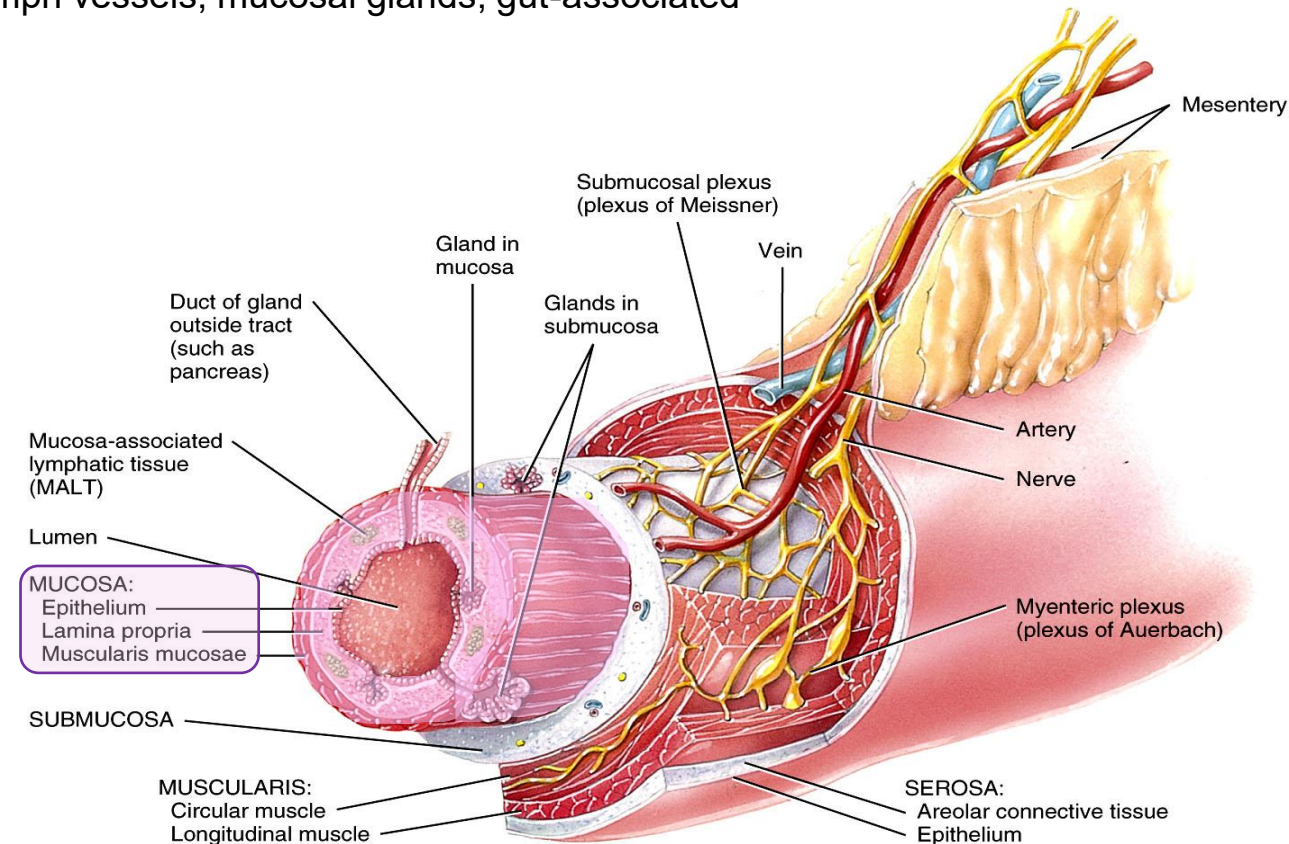


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Let's take a closer look at the Mucosa

The mucosa is in immediate contact with the food in the lumen. Its three main functions are:

- **Absorption** of digested nutrients, water & electrolytes into blood and lymph vessels
- **Secretion** of mucus, digestive enzymes, hormones & antibodies by various glands
- **Protection**
 - Physical barrier by separating the lumen (continuous with the external environment) from tissues and organs of the body
 - Immunological protection by lymphatic tissue serves the body's first line of immune defense

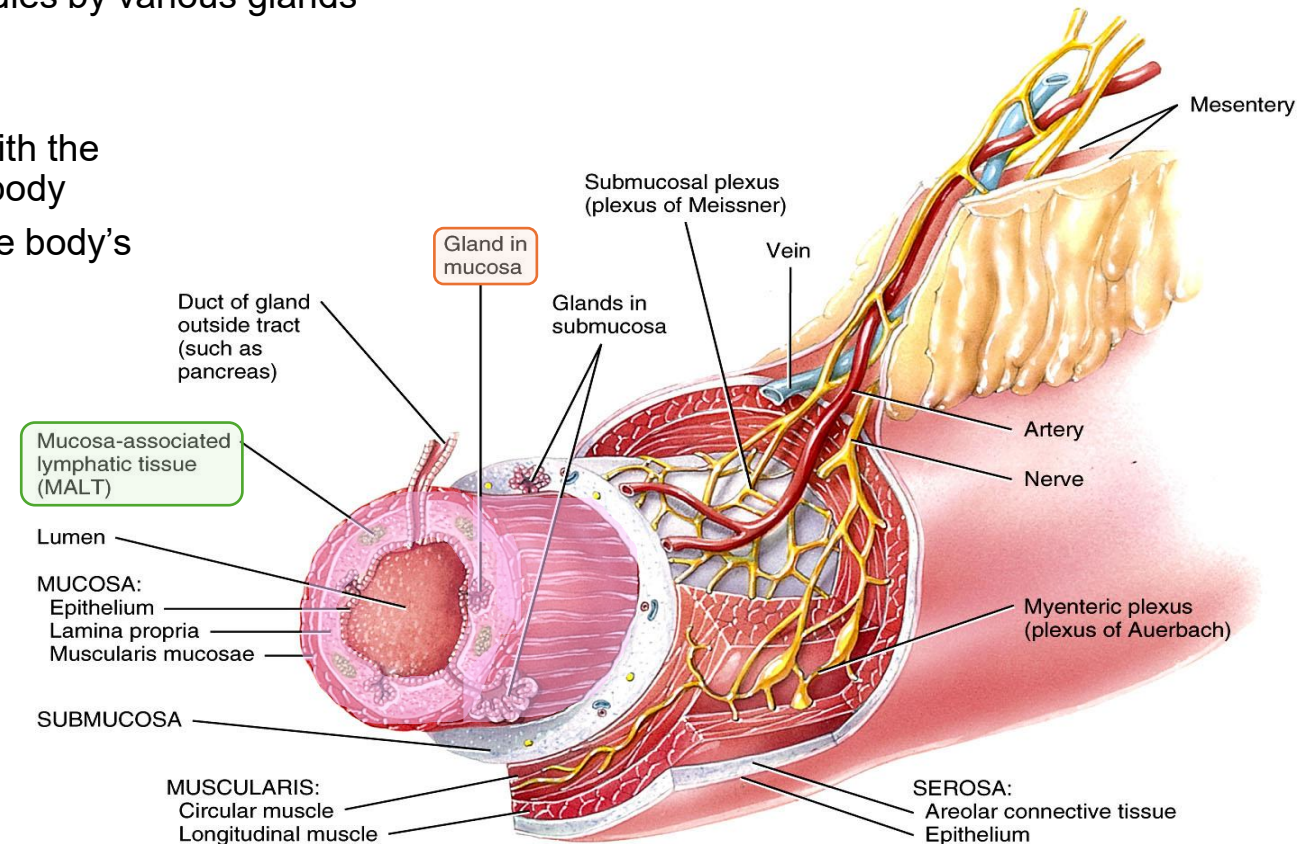


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Deep to the mucosa is the Submucosa

The submucosa lies between the muscularis mucosae and the muscularis externa.

Typical components includes:

- Dense irregular connective tissue
- Large blood vessels and lymphatic vessels → **site of esophageal and anal varices**
- Some areas contain submucosal glands: **Look out for these sites in your live lecture!**
- Contains a part of the enteric nervous system → **Meisner's submucosal plexus**

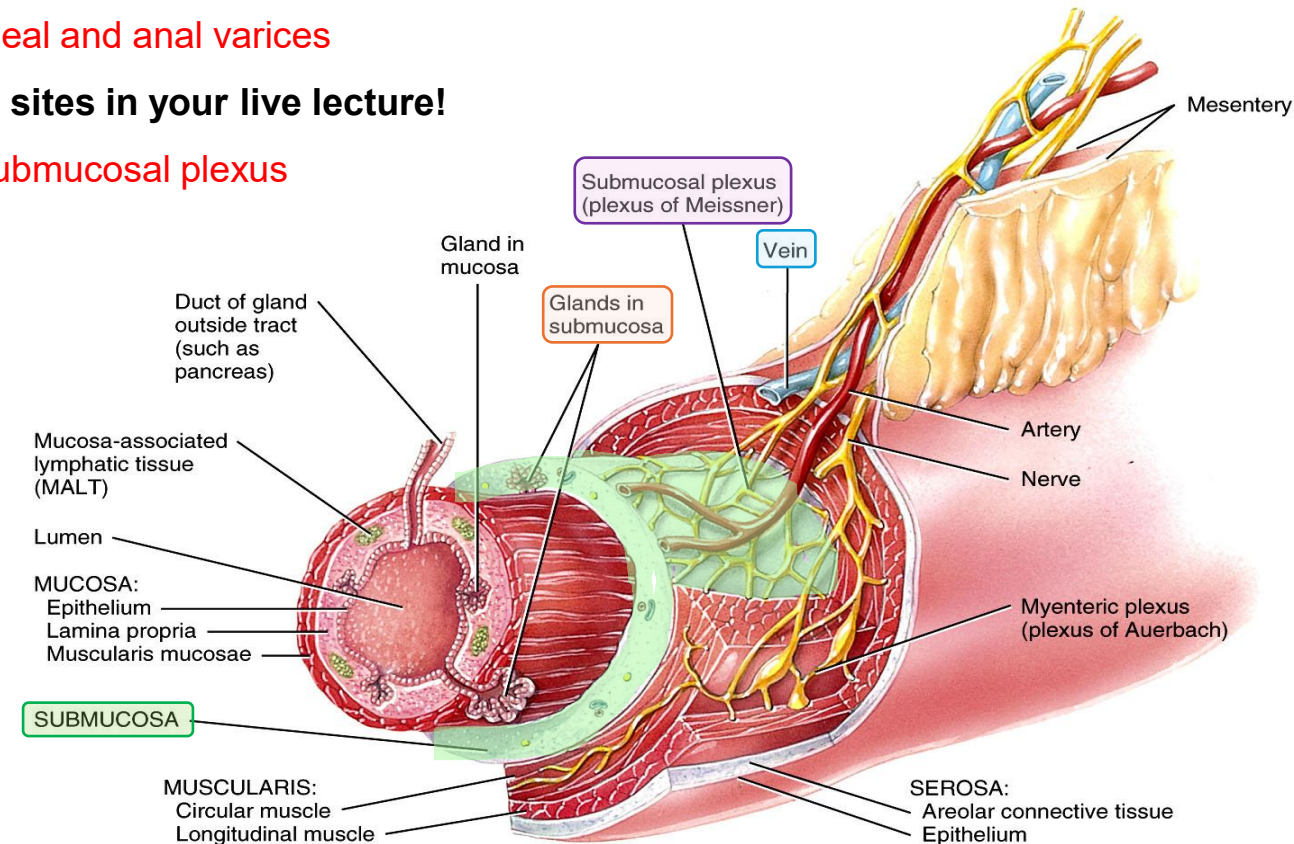
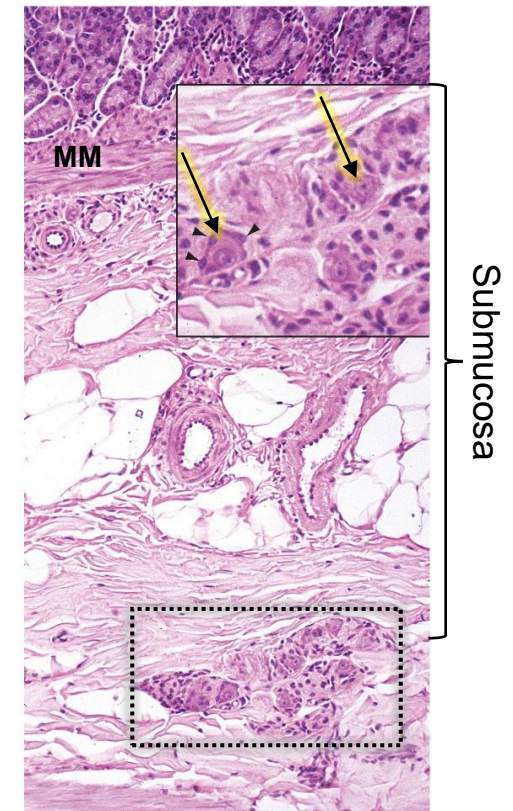
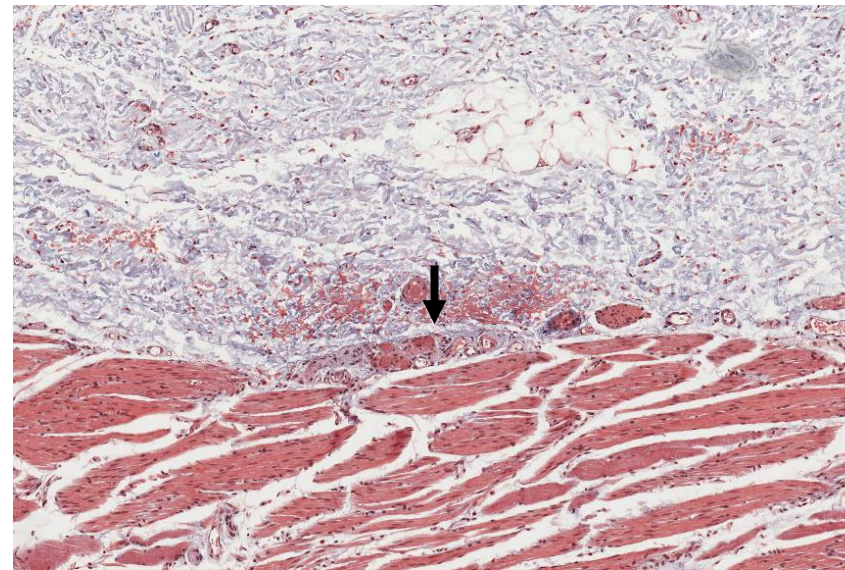
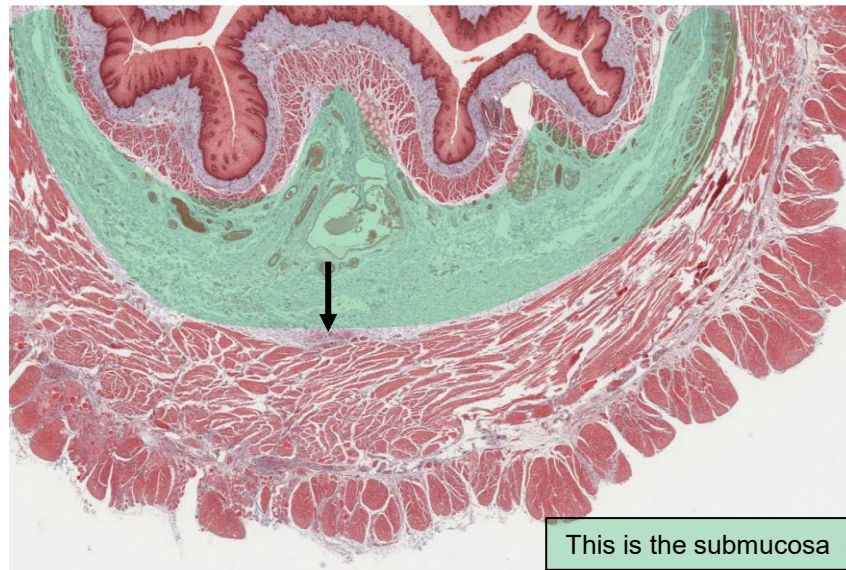


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Submucosal Plexus (Meissner's Plexus)

- Submucosal Plexus (Meissner's Plexus) can be identified as clusters of neuronal cell bodies located in submucosa → arrows
- They are the cell bodies of **postganglionic parasympathetic neurons** and innervate muscularis mucosae (MM) and other mucosal components



Task: What do you think is the embryological precursor to this structure?

Deep to the submucosa is the Muscularis Externa

The muscularis externa or simply muscularis typically contains two thick layers of smooth muscle

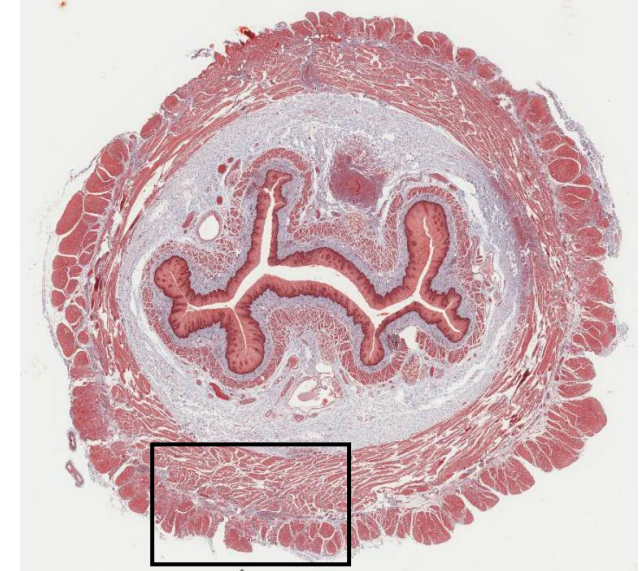
1. Inner Circular

- Contraction compresses and mixes content by constricting the lumen
- Forms sphincters at specific locations:
- Superior esophageal sphincter (Pharyngo-esophageal sphincter)
- Inferior esophageal sphincter
- Pyloric sphincter
- Ileocecal valve
- Internal anal sphincter
- These sphincters may be either the result of thickening of the circular layer or due to an increase in muscular tonicity

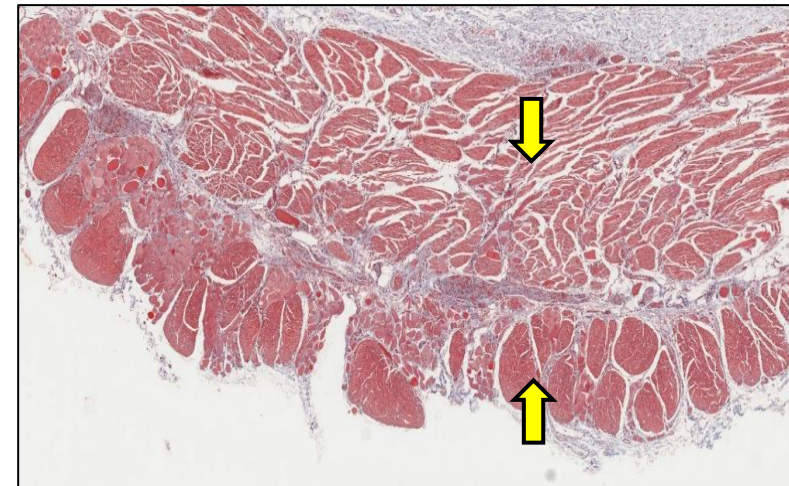
2. Outer Longitudinal

- Contractions propels content of lumen forward by shortening the tube

3. Contraction of these muscle layers produces **peristalsis (waves of contraction)** which moves content along the GI tract

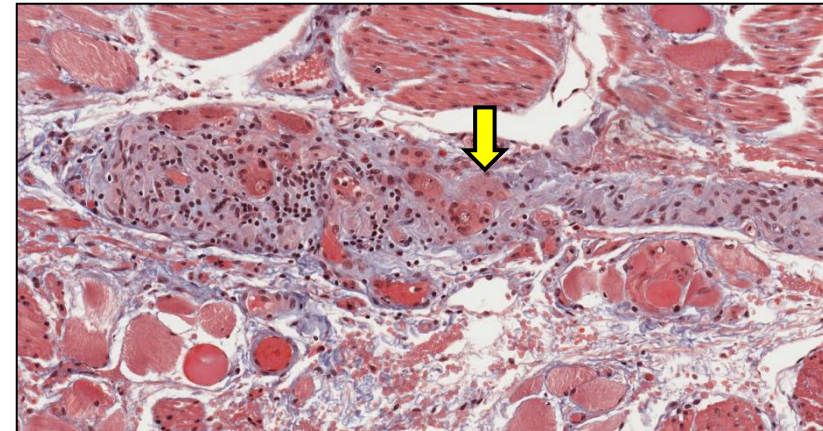
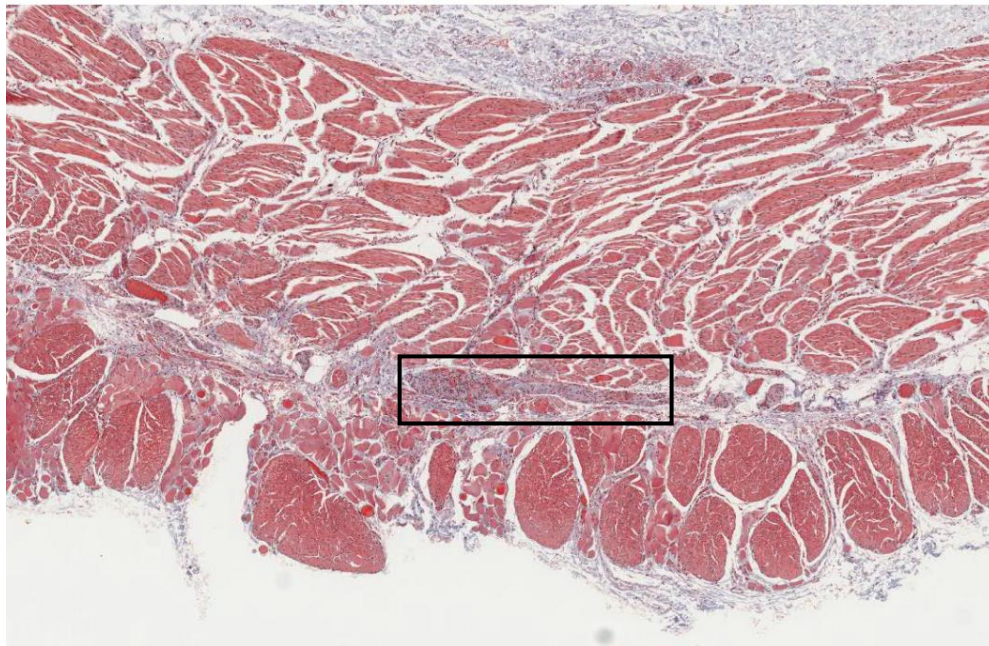


Let's take a closer look



Myenteric Plexus (Auerbach's Plexus)

- The Myenteric Plexus or Auerbach's Plexus also forms part of the enteric nervous system
- **It is always located between the circular and longitudinal muscle layers of the muscularis externa**
- Like the Meisner's plexus it contains **postganglionic parasympathetic neurons**
- It innervates the muscularis externa for slow, rhythmic contraction (peristalsis)



Pale staining parasympathetic ganglion cells located between inner circular (CM) and outer longitudinal (LM) muscle layers

Adventitia & Serosa

Figure A: This slide is a cross section through the GI Tract

- Serosa or serous membrane is connective tissue **lined by simple squamous epithelium or mesothelium**
- Mesothelium produces a small amount of fluid which allows friction free sliding of abdominal organs over each other once the serosa is intact → injury to this can result in adhesions
- **Organs with a complete serous lining (in the GI referred to as visceral peritoneum) are referred to as intraperitoneal and are suspended by a mesentery (double layer of peritoneum)**
- Large blood vessels, nerves and lymphatics travel through the mesentery and serosa to reach the inner layers of the wall of the digestive tract

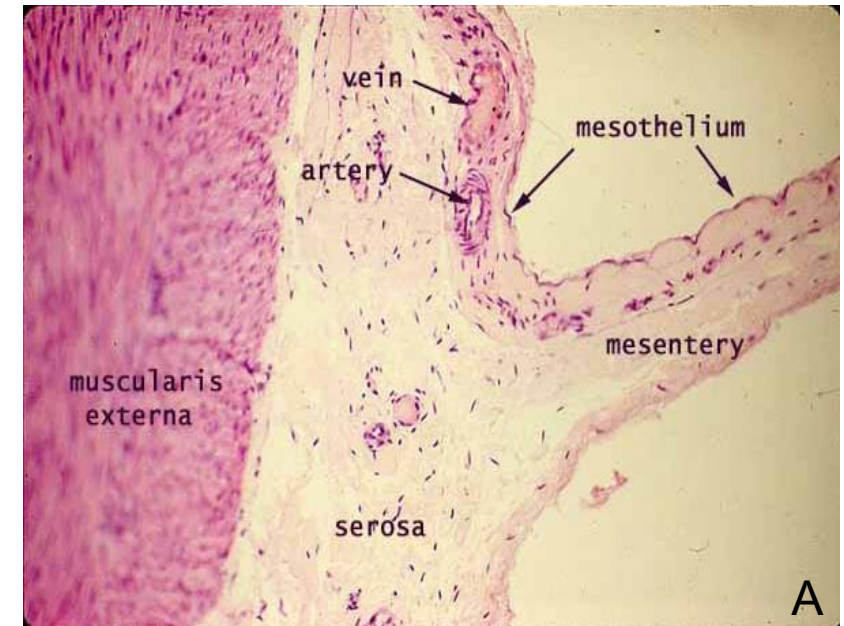


Figure B: This slide is a cross section through the superior mediastinum

- Adventitia is the external connective tissue covering of an organ or blood vessel. It protects and attaches structures to the body walls. **Parts of the GI tract that do not possess a complete serosal covering are called retroperitoneal**
- Observe the connective tissue fascia connecting the esophagus and trachea in Figure B

